Rodrigo Marques

+1 412 608 2487 • rodrigom@cs.cmu.edu
marques-rodrigo.github.io • rodrigo-marques2
marques-rodrigo • 0000-0003-2492-0197

Research Interests

Programming Languages, Type Inference, Substructural and Modal Type Systems

Education

Carnegie Mellon University & University of Porto — **Pittsburgh, PA & Porto, Portugal**
PhD in Computer Science — *Aug. 2025 – present*
– Studying type inference for substructural adjoint types.
– Advised by Frank Pfenning and Mário Florido.

Universidade do Porto — **Porto, Portugal**
MSc in Computer Science (Reliable Computing) — *Sep. 2019 – Dec. 2023*
– **Dissertation**: Subtyping: Study and Implementation.
– Studied type inference for algebraic subtyping with row polymorphism.
– Advised by Mário Florido and Pedro Vasconcelos.

Universidade do Porto — **Porto, Portugal**
BSc in Computer Science — *Sep. 2016 – Jul. 2019*

Publications

2024 **Haskelite: A Tracing Interpreter Based on a Pattern-Matching Calculus**
Pedro Vasconcelos, ***Rodrigo Marques***
Proceedings of the 17th ACM SIGPLAN International Haskell Symposium, pp. 1–13, ACM. doi:10.1145/3677999.3678274

Presentations

2023 **A Lazy Abstract Machine Based on a Pattern-Matching Calculus**
Pedro Vasconcelos, ***Rodrigo Marques***
Presented at the 35th Symposium on Implementation and Application of Functional Languages.

2022 **Towards Algebraic Subtyping for Extensible Records**
Rodrigo Marques*, Mário Florido, Pedro Vasconcelos*
Extended abstract presented at the ML Workshop, 27th ACM SIGPLAN International Conference on Functional Programming.

Teaching

Universidade do Porto — Lab Instructor

Spring 2025 **Theory of Computation** *(Formal Languages, Automata, Turing Machines)*
Fall 2024 **Functional and Logic Programming** *(Haskell, Prolog)*
Spring 2024 **Functional Programming** *(Haskell)*
Programming *(C++)*
Fall 2023 **Introduction to Programming** *(Python)*
Programming Fundamentals *(Python)*

Awards

2025 – 2030 **CMU Portugal Dual Degree PhD Fellowship**

Research Training

2026 **Summer School on Foundations of Programming and Software Systems**, ISCTE, Lisbon

2025 **Scottish Programming Languages & Verification Summer School**, University of Edinburgh

2024 **Oregon Programming Languages Summer School**, Boston University
International School on Rewriting, University of Innsbruck
International School on Proof Theory, University of Birmingham

2023 **EuroProofNet Summer School on Verification Technology, Systems & Applications**, INRIA Nancy

Academic Service

2026 **Student Volunteer**, Federated Logic Conference, Lisbon

2024 **Student Volunteer**, ACM SIGPLAN International Conference on Functional Programming, Milan

Professional Experience

Universidade do Porto — **Porto, Portugal**
Teaching Assistant — *Sep. 2023 – Sep. 2025*

NORTHWIND — **Montreal, Canada | Remote**
Gameplay Programmer — *Aug. 2021 – Aug. 2022*
– Developed and maintained gameplay systems for an online role-playing game in a remote team.

ROBLOX — **San Mateo, CA | Remote**
Accelerator Program Intern — *Summer 2021*
– Developed an ocean simulation and sailing mechanics for an online role-playing game.
– Introduced Git-based version control and integrated PlayFab analytics for project telemetry.

Skills

Programming: C++ · Haskell · Lua · Prolog · Python

Natural Languages: Portuguese (Native) · English (Fluent)